

Measuring the Central Tendency: (1) Mean

- Mean (algebraic measure) (sample vs. population):

Note: n is sample size and N is population size.


$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$


$$\mu = \frac{\sum x}{N}$$

- Weighted arithmetic mean:


$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

- Trimmed mean:

- Chopping extreme values (e.g., Olympics gymnastics score computation)